प्रदेश लोक सेवा आयोग, गण्डकी प्रदेश,
प्रदेश/ स्थानीय इन्जिनियरिङ्ग सेवा, सिभिल समूह, अधिकृत सातौं तह, इन्जिनियर पदको अन्तरतह, खुला प्रतियोगितात्मक लिखित परीक्षा
मिति: २०८१/०९/१०

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विषय: सेवा सम्बन्धी

Section-A (25 Marks)

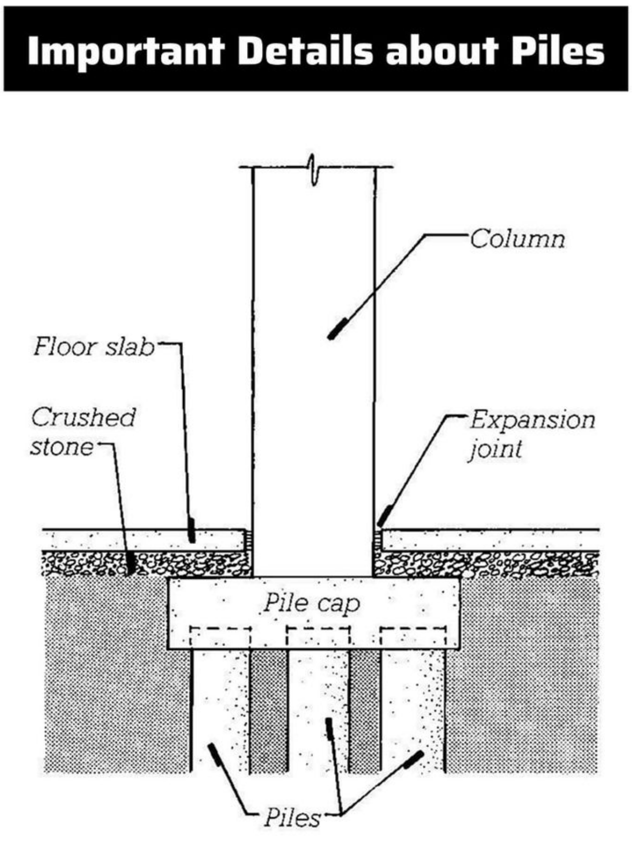
1. What is the difference between pile foundation and well foundation? Illustrate in figures. [5]

Answer:

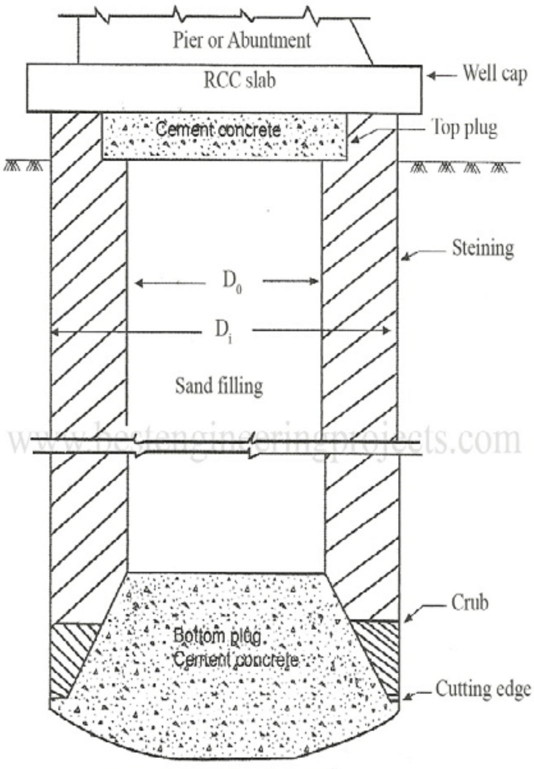
Both Pile and well foundation are the type of deep foundations. Those foundation whose depth is much larger than its width, is known as deep foundations.

Pile foundation is preferred for buildings and bridges when the upper soil is weak and loads need to be transferred deeper. Well foundation is preferred for bridge piers in rivers, where strong resistance to lateral forces and scour is needed.

	Pile foundation		Well foundation
1.	It consists of long, slender members to transmit loads to a deeper soil stratum.	1.	It consists of watertight structure used as bridge pier, construction of dam.
2.	In pile foundation, load is transferred by the end bearing, skin friction or both.	2.	In well foundation, load is transferred by base resistance and side resistance after sinking.
3.	The pile is driven or bored into the ground.	3.	The well is gradually sunk by excavation under the well.
4.	Construction of pile foundation is usually faster and cheaper.	4.	Construction of well foundation is slower and costlier.
5.	Pile foundation is easier to sunk to the required depth.	5.	Well foundation is sunk to the required level below the scour depth.
6.	Pile foundation is used in buildings and bridges when the bearing capacity of soil near the surface is relatively low.	6.	Pile foundation is used in bridges piers, dam etc. and suitable for heavy lateral loads.



a) Pile Foundation



b) Well Foundation

2. **Discuss the various factors that affects permeability of the soil. [5]**

Answer:

Permeability of a soil is the property of soil that allows the water to pass through the voids present in it. It can be defined as the easiness with which water can flow through the pores of soil mass. Permeability property of a soil results in seepage ultimately causing water loss.

According to Darcy's law, the velocity of water is directly proportional to hydraulic gradient (i) in a pipe flow such that:

$$v = k * I$$

where k = coefficient of permeability of soil or hydraulic conductivity.

The factors affecting permeability of soil are described below:

a. Void ratio:
- Larger the void ratio, more porous the soil mass which increases the permeability of soil.
- $k \, \alpha \, e^3 / (1+e)$

b. Size of soil particles:
- Coarser the soil particles, larger the size of voids and higher permeability.
- According to Hazen's equation:

$K = CD_{10}{}^2$ where C= constant

D_{10} = effective size of particle

c. Shape of soil particles:
- Rounded soil particles have lower specific area and higher permeability while angular soil particles have high specific area and lesser permeability

d. Viscosity of water:
- Viscous liquid reduces the permeability of soil.
- $k \propto 1/\mu$

e. Temperature:
- As the temperature increases, viscosity of liquid decreases which increases the permeability of soil.
- $k \propto 1/\eta$

3. **Differentiate between consolidation and compaction of soils. What are the factors that affect the compaction? Briefly describe the three stages of consolidation of soil. [4+3+3=10]**

Answer:

Soil compaction is the process of densification of soil by removing the air voids present in the soil particles under the action of mechanical energy. Similarly, consolidation is the process of compression of water voids due to which expulsion of water from the void occurs under the effect of applied load.

The differences between soil compaction and consolidation are given in points.

	Compaction		Consolidation
1.	Densification of soil under the action of mechanical energy.	1.	Compression of water voids under the effect of sustained applied load.
2.	Compaction is immediate process.	2.	Consolidation is slow and time-dependent process.
3.	It occurs due to reduction of air voids.	3.	It occurs due to reduction of water filled voids.
4.	Compaction increases dry density and strength of soil.	4.	Consolidation causes settlement and increase in effective stress.

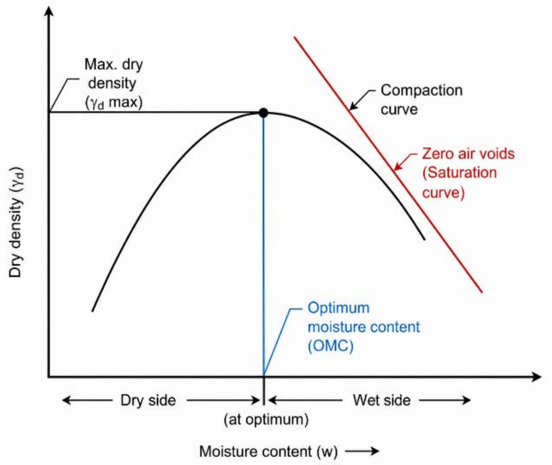
5.	Compaction is generally artificial and intentional process.	5.	Consolidation is generally natural or load-induced.
6.	Equipment such as roller, rammer, vibrators are used for compaction.	6.	No mechanical equipment is required for consolidation. It occurs under structural load.

Factors affecting compaction:

There are several factors affecting compaction. They are described in brief:

i. Moisture content of soil:

Maximum dry density of the soil is achieved at Optimum Moisture Content (OMC). Decrease in moisture content than OMC, makes it dry of optimum compaction while increase in moisture content than OMC makes it wet of optimum compaction.



ii. Amount of energy applied:

Larger maximum dry density can be achieved even at lower OMC, by application of larger magnitude of energy.

iii. Types of soil:

Coarser type of soil achieves higher dry density than finer soil type.

Consolidation is the long-term process in which removal of water from the voids occurs under the sustained applied load. The primary effect of consolidation is occurrence of ground settlement. If the settlement occurs uniformly, the structure will settle slightly without major structural damage. However, differential settlement across a foundation causes different parts of foundation to sink at different rates, leading to severe twisting, structural cracking and foundational instability.

There are **three stages of consolidation**. They are briefly described as follows:

a. Initial Consolidation:

Consolidation that occurs immediately after loading due to compression and expulsion of air from voids in partially saturated soil is known as initial consolidation. However, the consolidation is very small in fully saturated clays.

Formulae:

Immediate settlement:

$$S_i = \frac{qB(1-\mu^2)}{E_s} \times I_s$$

$$where:$$

$$\mathrm{S_i} = \text{Immediate settlement(m)}$$
$$\mathrm{q} = \text{Applied pressure}(\mathrm{kN/m^2})$$
$$\mathrm{B} = \text{Width of footing(m)}$$

E_s = Modulus of elasticity of soil(kN/m^2)
μ = Poisson's ratio
I_s = Influence factor

b. **Primary consolidation:**

Consolidation that occurs when excess pore water is gradually expelled from soil voids is known as primary consolidation. It starts after the initial consolidation and is the most significant stages. As the water expelled out from the void, the settlement takes place which may take several days to year.

Settlement formula:

For normally consolidated clay:

$$S_p = \frac{C_c H}{1+e_0}\log_{10}\left(\frac{\sigma_0' + \Delta\sigma}{\sigma_0'}\right)$$

For over-consolidated clay:

$$If\ \sigma_0' + \Delta\sigma \le \sigma_c':$$

$$S_p = \frac{C_r H}{1+e_0}\log_{10}\left(\frac{\sigma_0' + \Delta\sigma}{\sigma_0'}\right)$$

$$If\ \sigma_0' + \Delta\sigma > \sigma_c':$$

$$S_p = \frac{C_r H}{1+e_0}\log_{10}\left(\frac{\sigma_c'}{\sigma_0'}\right) + \frac{C_c H}{1+e_0}\log_{10}\left(\frac{\sigma_0' + \Delta\sigma}{\sigma_c'}\right)$$

where:

Sp = Primary consolidation settlement
Cc = Compression index
Cr = Recompression index
H = Thickness of clay layer
e_0 = Initial void ratio
σ_0' = Initial effective stress
$\Delta\sigma$ = Stress increment

c. **Secondary consolidation:**

Consolidation in which occurs due to rearrangement of the soil particles after the completion of primary consolidation is known as secondary consolidation. It starts after all the excess pore water pressure has dissipated. The consolidation can continue for several years.

Formulae:

$$S_s = \frac{C_\alpha H}{1+e_p}\log_{10}\left(\frac{t_2}{t_1}\right) Ss$$

where:

S_s = Secondary settlement
$C\alpha$ = Secondary compression index
H = Thickness of compressible layer
e_p = Void ratio at the end of primary consolidation
t_1 = Time at end of primary consolidation
t_2 = Time after t_1

4. **List the loads, forces and stresses to be considered in designing a highway bridge. What is the impact factor? State formula for calculating impact factor for RCC bridge. [6+2+2=10]**

Answer:

Various loads and forces acting on the bridges are discussed below:

a. Live load:

- Moving load along the length of the road is known as live load.
- According to IRC, live load can be:
  - **i. IRC Class A loading:**
    - Treated as standard loading which is adopted for the design and construction of permanent bridges.
  - **ii. IRC Class B loading:**
    - Adopted for the design and construction of temporary bridges and bridges in specifies areas.
  - **iii. IRC Class AA loading:**
    - Treated as abnormal loading used for construction of bridges where heavy traffic load is to be subjected.
  - **iv. IRC Class 70R loading:**
    - Treated as abnormal loading, which has three patterns of loading i.e. wheeled load, tracked load and train load.

b. Impact load:

- Dynamic effect of moving live load along the length of bridge.
- Impact load is determined by using the formula:

Impact load= Impact Factor x Static value of live load

c. Wind load:

- Stress subjected due to wind on the exposed part of bridge.
- Wind load= Wind load on the structure + Wind load on live load.
- $F_t = P_z \times A \times G \times C_D$

Where F_t = Wind load in transverse directions
P_z = Design wind pressure
A= Area of the exposed part of structure
G= Gust Factor
C_D= Coefficient of Drag

d. Longitudinal forces:

- Horizontal forces acting parallel to the length of road.
- Longitudinal forces on the bridge occurs:
  - i. Due to tracking effect through acceleration of driving wheels.
  - ii. Due to braking effect through application of brakes.
  - iii. Frictional resistance offered to the movement of free bearing.

e. Lateral forces:

- In case of highway bridge, lateral load due to parapet and kerb is normally considered.
  - i. Parapet:
    Shall be designed to resist lateral horizontal forces and vertical forces each of 150 kg/m applied simultaneously on the top.
  - ii. Kerbs:
    Shall be designed for lateral loadings of 750 kg/m applied horizontally on the top of kerbs.

Impact factor is the factor considered in bridge live load to account for the dynamic effects of moving vehicles, such as vibration and sudden wheel loads. The factor increases the static load so that bridge is safe under traffic conditions as:

Impact load= Impact Factor x Static value of live load

Formula of impact factor for RCC bridge:

For **RCC bridges**, the impact factor is commonly calculated using the formula:

$$\boxed{IF = \frac{4.5}{6 + L}}$$

Where, IF= Impact factor

L= effective span of the bridge (m)

The **impact load** is then:

$$\boxed{\text{Impact Load} = IF \times \text{Live Load}}$$

The **total design live load** becomes:

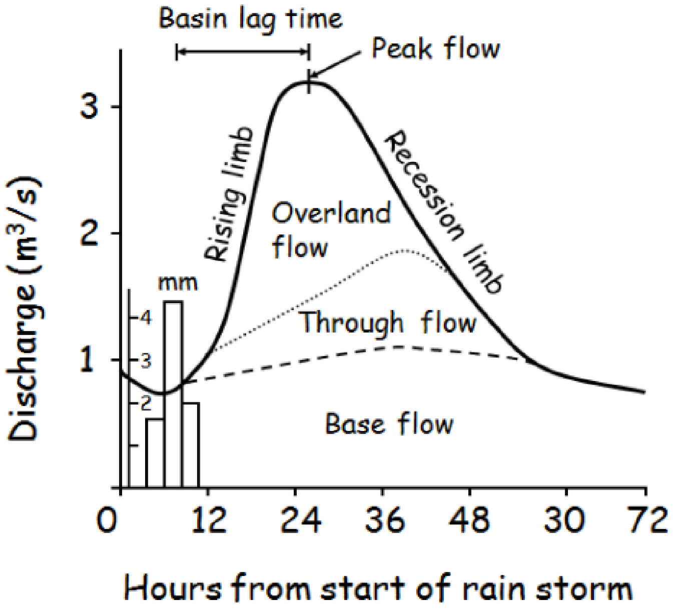
$$\boxed{\text{Design Live Load} = \text{Live Load}\,(1 + IF)}$$

Section-B (25 Marks)

5. Discuss the principles and methods of hydrograph analysis and its importance in flood frequency analysis. [5]

Answer:

A hydrograph is the response of a given catchment to a rainfall input. It consists of flow in all the three phases of runoff. It consists of surface runoff, interflow and base flow. Two rainfall in a given catchment produce different hydrographs. Similarly, hydrograph analysis is the study of how stream discharge changes with time after rainfall.



Principles of hydrograph analysis:

i. A hydrograph reflects the catchment response to a storm, so its shape depends on rainfall and basin characteristics.

ii. The total hydrograph is usually treated as the sum of direct runoff and base flow.

iii. The hydrograph response is not instantaneous because water moves through the catchment over time. That delay produces the rising limb, peak flow, and recession limb.

Methods of hydrograph analysis:

i. Baseflow separation:

This method separates the groundwater flow from the direct runoff. There are various methods of baseflow separation. They are: straight line method, variable slope method etc.

ii. Unit Hydrograph method:
This method uses the unit hydrograph which the direct runoff hydrograph resulting from the unit depth of excess rainfall occuring unifromly over the basin at uniform rate for specific duration. It is used to estimate flood hydrographs for different storms.

iii. Flood hydrograph analysis:
This method of hydrograph analysis determines the important characteristics like peak discharge, base time, rising limb, falling limb and total runoff volume.

iv. Instantaneous unit hydrograph:
This method if used when observed runoff data are unavailable. These hydrographs are developed from watershed characteristics such as catchment area, length of streams, time of concentration and so on.

Importance of hydrograph anlaysis:

i. It helps in determination of peak flood discharge used in flood frequency studies.
ii. Hydrograph analysis also helps in design of hydraulic structures such as dams and spillways.
iii. It also helps in flood risk assessment and disaster management.
iv. The analysis also helps in reservoir design and operation and flood routing.
v. It also helps in flood forecasting and warning.

6. Explain the specific consideration to be taken for the design, operation and management of hill irrigation system. [10]

Answer:

Steep slopes, rugged topography, unstable geology and scattered agricultural land makes the hill irrigation systems different from those in plains. Hence, specific considerations need to be taken for the design, operation and management of hill irrigation system. They are listed below:

i. Design considerations:
- Reliable source of water must be selected for the withdrawing adequate amount of discharge for irrigation systems.
- Canals need to be aligned along the contour lines to minimize excavation and erosion.
- Landslide prone areas, unstable slopes and excessive cutting of hillslopes must be avoided for construction of irrigation system.
- Permissible flow velocity must be maintained in the irrigation systems to prevent erosion or sediment deposition.

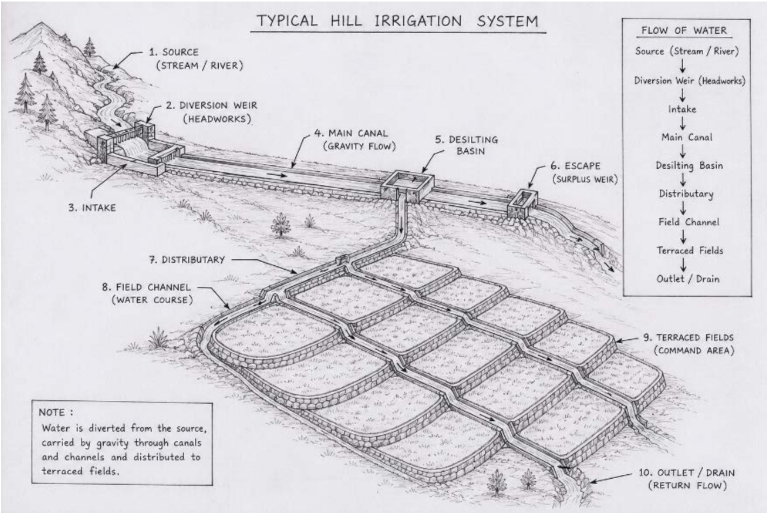
ii. Managerial and Institutional considerations:
- Regular inspections and maintenance work of canals, diversion structures, retaining walls etc. are required.
- Regulating structures must be easy to use for the local farmers to operate and repair.

iii. Agricultural considerations:
- The irrigation system design must be compatible with the soil type of the terrain and the crops that are grown in hilly terrain.
- Water must be designed to be delivered in a way that are suitable to the local farming pattern.

iv. Financial considerations:
- Design of the irrigation systems in the hill terrain must be economical and affordable.
- Local labor and locally available materials should be used for construction, maintenance and repairs to improve long term sustainability and economical aspect.

v. Social arrangements:

- Irrigation systems must be constructed by protecting the water rights, local customs and ethnic groups.
- The construction of irrigation system should not instigate any conflict and make water distribution acceptable to users.



7. **Discuss the challenges and solution associated with integrating hydropower into the modern energy grid, particularly with respect to grid stability, load balancing and energy storage. How can advances in smart grid technologies aid in optimizing hydropower generation? Also, discuss irrigation pattern during hydropower planning. [4+4+2=10]**

Answer:

Hydropower is the important renewable energy sources as it provides clean, flexible energy and capable of responding quickly to changes in electricity demand**.** However, integrating hydropower into a modern power grid presents several technical, environmental, and operational challenges.

i. Challenges related to grid stability:

Grid stability refers to the ability of the electrical system to maintain constant voltage and frequency despite changes in generation or demand.

a. Seasonal river flow variation:

- During dry season, river discharge decreases, reducing the electricity generation. Similarly, in monsoon season, excess water may exceed the generation capacity.
- It may lead to fluctuating power supply and frequency instability which can be solved by construction of reservoir-based hydropower plants and integrate multiple renewable sources.

b. Frequency stability:

- It refers to the sudden change in electricity generation or demand that disturbs the system frequency.
- It leads to load fluctuations, sudden generator outages and variable renewable energy generation.
- The challenge can be solved by integrating governor control systems, Automatic Generation Control (AGC) and grid frequency monitoring.

c. Voltage stability:

- Long transmission lines from remote hydropower plants can cause voltage fluctuations.
- The challenge can be solved by upgrading transmission infrastructures, reactive power infrastructure and providing Static VAR Compensators (SVC)

d. Grid synchronization:

- Connecting many hydropower stations requires precise synchronization of voltage, frequency and phase angles.
- The challenge can be solved by integrating with advanced protection systems, smart grid technologies and digital monitoring and control systems.

ii. **Challenges related to load balancing:**

Load balancing means matching electricity generation with consumer demand at all times.

a. **Daily Demand Variation:**
   - The demand of electricity changes throughout the day.
   - The challenge can be solved by inclusion of flexible turbine operation, peak-load hydropower plants and demand-side management.

b. **Transmission Constraints:**
   - Hydropower plants located in mountainous and remote regions creates the challenge of long-distance transmission, transmission losses and grid congestion.
   - The challenge can be solved by grid expansion, regional power connections and provision of high voltage transmission lines.

iii. **Challenges related to energy storage:**

a. **Limited water storage:**
   - ROR projects have little to no storage capacity.
   - It cannot store excess energy and unable to generate up to the required demand during dry seasons.
   - Hence, reservoir-based hydropower, cascade hydropower systems and multipurpose dams must be given more importance.

b. **Pumped storage hydropower:**
   - This is the most effective large-scale energy storage technology.
   - It helps to store surplus renewable energy, supports frequency regulation and improved grid reliability.
   - However, the construction cost of such hydropower project is high and requires suitable topography for construction.

Smart grids are the modern electricity networks that use digital communication, sensors, automation and advanced control systems to monitor and manage the electricity generation, transmission, distribution and consumption of electricity in real time. They help in improving the efficiency, reliability and flexibility of hydropower generation.

i. **Real time monitoring and control:**
   - Smart grids continuously monitor the water level in reservoirs, river flow rates, turbine performances, grid frequency and voltage.
   - It helps in optimizing power generation according to water availability.
   - It also detects the equipment faults at early and reduces outage by improving plants efficiency.

ii. **Load forecasting and demand prediction:**
   - The real time data as well as previous data can be analyzed by using AI and Machine learning for predicting electricity demand.
   - Hydropower plants can schedule energy generation as per the required demand and improved energy efficiency.

iii. **Integration with renewable energy sources:**
   - Smart grids help in proper integration with the renewable energy sourced like solar, wind and batteries which ensures continuous and reliable power supply.

- Hydropower quickly compensated for the fluctuations in solar and wind energy generations.

iv. Demand response management:

- Smart meters and communication systems allow consumers to adjust the electricity usage during peak demand periods.
- It reduces peak load and balances electricity demand and supply.

v. Remote Automation (SCADA)

- SCADA (Supervisory Control and Data Acquisition) systems enables remote monitoring and control of hydropower plants.
- It helps in faster operational decisions and efficient plant operations by reducing manpower requirements.

Irrigation pattern refers to the seasonal and spatial distribution of irrigation water demand in the command area served by a river or reservoir. Since both hydropower generation and irrigation require the same water resource, understanding the irrigation pattern is essential for planning multipurpose projects and avoiding conflicts between energy production and agricultural water supply. **According to the Water Resources Act 2049**, the priority order for utilizing natural resources should be as:

a. Drinking water and domestic users
b. Irrigation
c. Agricultural use
d. Hydroelectricity
e. Industries
f. Navigation
g. Recreation
h. Other uses

The key irrigation pattern considerations in hydropower planning:

i. Crop water requirement:

Different crops have different water requirement.

Crop	Water Demand
Rice (Paddy)	Very High
Wheat	Moderate
Maize	Moderate
Vegetables	Variable
Sugarcane	High

ii. Seasonal Demand Variation:

In Nepal,

- Monsoon (June–September):
  River flow is high; irrigation demand for paddy is also high.
- Winter (November–February):
  River flow decreases, but wheat irrigation is needed.
- Pre-monsoon (March–May):
  Water scarcity is often severe.

Hence, this creates a conflict between power generation and irrigation.

Section-C (25 Marks)

8. Describe with sketches the elements of highways cross sections. [5]

Answer:
Highway cross-section is the section of road cut perpendicular to the centerline axis of the road. The elements of highway cross-section influence the life of the pavement as well as riding comfort and safety. It consists of many parts such as carriageway, shoulder, median strips, traffic separators, side drain, side slope, catch drains etc.

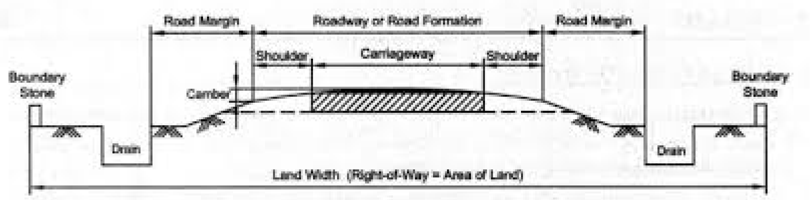


FIG. 1 CROSS-SECTION DETAILS OF A ROAD

i. **Carriageway:**
- It is the strip of road meant for vehicular traffic movement also known as pavement width.
- Width of carriageway depends upon the width of traffic lanes and number of lanes.
- According to NRS 2070, The minimum lane width for a single lane road is 3.75 m.
- Two land road is provided with minimum of 3.5 m for each lane.

ii. **Shoulder:**
- They are the strips provided along the road edge and are intended for the accumulation of stopped vehicles.
- Shoulders serve as emergency lane for vehicles and also provide the lateral support for base and surface courses.
- According to NRS 2070, the width of shoulders on either side of carriageway shall be at least 0.75m.
- It is desirable that color and texture of shoulders be different from those of the carriageway.

iii. **Roadway width:**
The sum of width of pavements or carriageway including separators and shoulders is called width of formation or roadway width.
It is the top width of highway embankment or bottom width of highway cutting excluding the side drains.

iv. **Side slope:**
- According to NRS 2070, side slopes of embankment and cuttings depend upon the type of fill/ cut materials and height/depth of fill or cut.
- If natural cross slope of the ground is more than 1:5 then the ground should be cut with more than 2 m wide horizontal steps.

v. **Lay-bys, bus bays and bus stops:**
- Lay-bys are the intermittent shoulders sufficiently wide and long provided to meet the important function of shoulder where the continuous shoulder on either side cannot be provided from economic considerations.
- Bus-bays shouldn't be located too close near the intersections. It is desirable that they are located at least 40 m from the intersection on either side.
- Bus stops should be placed adjacent to the bus linear line of travel so that it doesn't need to pull over to left. It must be provided at the spacings of 200-00 m.

vi. **Right of way:**
It is the width of land to be acquired for the road along its alignment. It must be adequate to accommodate all the cross-sectional elements of highway and reasonably provided for future development.
According to NRS2070,

ROW requirements for national highways= 50 m (25 m on either side of the road centerline)
Feeder roads = 30 m (15 m on either side of road centerline)
District roads= 20 m (10 m on either side of road centerline)

9. **Explain in details about the causes of failures of rigid pavements in highways. What are the methods to be adopted for the maintenance of rigid pavement roads? Also, explain why the highway maintenance is a challenging job to the engineering professionals in Nepal. [4+4+2=10]**

Answer:

Rigid pavements are those which posses' considerable flexural strength, made up of cement concrete which may either be plain, reinforced or prestressed. They have a slab action and are capable of transmitting the loads stresses through a wider area below.

There **are several causes of failures of rigid pavements** in highways. They are:

i. **Overloading and repeated traffic loading:**
- Heavy axle loads exceeding the design capacity produce high tensile stresses.
- Repeated wheel loads cause fatigue failure of concrete.
- It results in longitudinal cracks, transverse cracks.

ii. **Inadequate pavement thickness:**
- If the slab thickness of the pavement is insufficient, tensile stresses exceed the strength of concrete which reduces the pavement life significantly.

iii. **Weak or poorly compacted subgrade:**
- A weak subgrade causes uneven settlement, loss of support the slab and increases bending stress.
- It resulted in cracking, faulting, pumping and settlement.

iv. **Temperature stresses:**
- The expansion of concrete during hot weather and contraction during cold weather cases warping, curling, joint opening and thermal cracking.

v. **Moisture variation:**
- The change in moisture content causes differential expansion and contraction which gives rise to curling of the slabs, warping stresses, cracks and deterioration of the joints.

vi. **Improper joint design:**
- If the joints are designed too far, improperly sealed or poorly constructed, then the failures of rigid pavements such as random cracking, joint spalling, faulting etc. takes place.

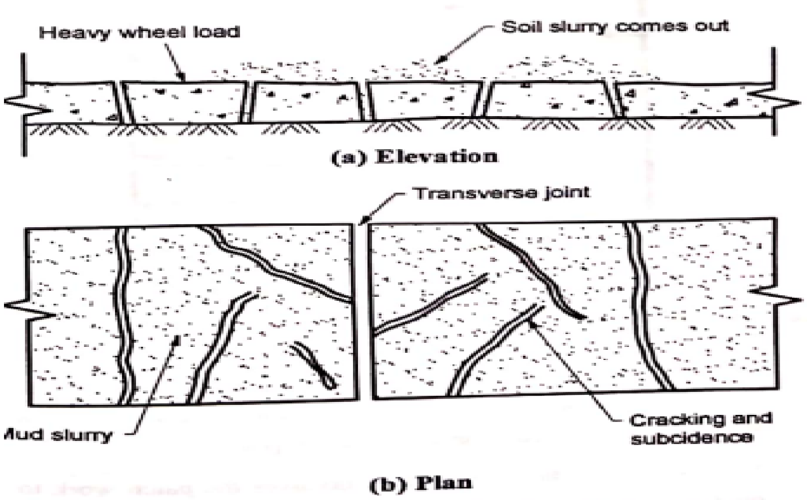


Fig. 7.2.10 : Mud pumping and failure of pavement

Methods to be adopted for the maintenance of rigid pavement roads:

i. **Preventive maintenance:**
   - Joint sealing and resealing to stop water and dirt from entering joints.
   - Crack sealing and filling to prevent cracks from widening.
   - Routine cleaning of pavement, shoulders, drains etc.

ii. **Corrective maintenance:**
   - Partial-depth patching for surface defects such as spalling, scaling and small damaged areas.
   - Repair of settlement and pumping areas by stabilizing the support and restoring slab level.
   - Dowel bar retrofitting or joint repair to improve load transfer.

The difficult landscape, rugged topography, extreme weather conditions and limited resources makes the highway maintenance challenging job to the engineering professionals. The challenges for the highway maintenance are described below:

i. **Difficult topography:**
   Highway passing through unstable and hilly terrain makes the maintenance works difficult and expensive.

ii. **Landslide and slope instability:**
   Heavy rainfall might trigger landslides, rockfalls and slope failure. Hence, highway maintenance might be difficult and costly due to frequent blocked and damaged road.

iii. **Geological conditions:**
   Fragile geological landscapes in hilly terrain pose great challenges in highway maintenance works.

iv. **Remote and inaccessible locations:**
   Hill roads pass through isolated mountain regions with poor accessibility which makes it difficult to carry out highway maintenance works.

10. Why road maintenance is necessary? Describe the different types of road maintenance. Explain the maintenance of bituminous pavement. [10]

Answer:

Highway road maintenance is the task of preserving, repairing and restoring a system of roadways to keep the serviceable conditions of highway as normal as possible and best as practicable. It is importance as maintenance helps in providing better service facilities longer life and better appearance. It involves variety of planning, programming and scheduling to actual implementation in the field and monitoring.

Highway Maintenance activities can be classified into three categories:

i. **Routine maintenance activities:**
   - Those activities that are carried out as frequently as required during each year on all elements of the highway to improve its serviceability in all weather conditions are routine maintenance activities.
   - Includes:
     a. Grass and bush cutting, grading, reshaping of the unpaved surfaces.
     b. Cleaning of carriageway, drains, slopes, culverts.
     c. Replacement of structures that has been damaged.

ii. **Recurrent maintenance activities:**

- Those maintenance activities of localized nature carried out more or less at regular intervals of six months to 2 years are recurrent maintenance activities.
- Includes:

  a. Sealing cracks, local surface treatment

  b. Repair of depression and ruts

  c. local constructions

iii. **Periodic maintenance activities:**

- Those activities done at the interval of several years to improve the overall functionality of road are periodic maintenance activities.
- Includes:

  a. Renewal of wearing course of the pavement surface.

  b. Restoration of road markings

  c. Repainting of metal bridges etc.

iv. **Special maintenance activities:**

- Those activities of specialized and emergency nature done to solve special problems which may cause pavement failure if not identified on time.
- Includes:

  a. construction of temporary diversion to allow the traffic to pass through obstruction.

  b. Removal of debris and other obstacles.

  c. Strengthening of pavement structures.

Road maintenance can also be grouped as:

i. **Preventive maintenance:**

It included actions that are taken to prevent premature deterioration and to retard the progression of deficiencies to reduce the rate of deterioration and increases useful life of pavement.

ii. **Corrective maintenance:**

It includes the action that includes actions taken to correct the deficiencies which are hazardous which seriously affect a pavement's operation to keep the highway within a tolerable level of serviceability.

Maintenance of bituminous pavement:

i. **Routine maintenance:**
   It consists of minor works carried out regularly to prevent deterioration of pavement such as:
   a. filing potholes
   b. sealing cracks
   c. cleaning culvert
   d. removing debris from the shoulders
   e. vegetation control etc.

ii. **Preventive maintenance:**
   It is performed to prevent the severe damage of the bituminous pavement such as:
   a. Crack sealing
   b. Fog seal
   c. Slurry seal
   d. Micro surfacing
   e. Surface dressing etc.

iii. **Corrective maintenance:**
   It is performed to repair localized pavement failures after the damage has been occurred such as:
   a. Pothole repair
   b. Patch repairs
   c. Rut correction etc.

iv. **Periodic maintenance:**
   It is performed at regular intervals typically every 5-10 years to restore pavement conditions such as:
   a. Overlay
   b. Milling and overlay etc.

v. **Rehabilitation:**
   Rehabilitation of bituminous pavement is carried out when pavement deterioration becomes severe but reconstruction is not yet necessary. It includes:
   a. Structural overlays
   b. Base strengthening:
   c. Full depth reclamation etc.

vi. **Reconstruction:**
   When the pavement is beyond repair, existing pavement is completely removed and new pavement layers are constructed.

Section-D (25 Marks)

11. **As a site engineer how will you describe the process of pipe laying in water supply project? As a sanitary engineer, do you suggest for composting and incineration methods for sludge treatment in Nepal. Explaining strength and weakness of each method. [5+5=10]**

Answer:

Pipe laying is the process of installation of pipelines to transport water in water supply project. The process of pipe laying in the water supply project is briefly explained in points.

i. **Preparation of detailed maps:**
At the first step of laying of the pipe, survey is conducted to prepare the detailed map showing the alignment of pipelines from the source to treatment plane and distribution network.

ii. **Locating proposed alignment of pipeline on the ground:**
The alignment is marked on the ground by driving central stakes 30 m apart on the straight reaches and 7.5 m to 15 m apart on the curves on roads.

iii. **Location of the pipes with respect to the ground surface during laying:**
Pipes can be laid above the ground from the source to treatment plant and distribution mains or below the ground surface. If they are to be laid on the ground surface, they must be well compacted and supported by masonry or concrete support at 5 to 15 m. if they are to be laid below the ground surface, then trench must be excavated.

iv. **Excavation and preparation of trench:**
It can be done mechanically or manually as per specification. Trench should be excavated by adding 30 to 45 cm in external diameter of pipe. More than 90 cm clearance is provided above the crown but 15 to 20 cm more excavation is done in both directions at the place of joint.

v. **Dewatering of trench:**
When the GWT is high, dewatering of trench must be done with the help of pump or other means.

vi. **Lowering the pipes into the trench:**
The pipes in one side of trench are clear to remove external material and gently lowered into the trench so as not to damage the protective coatings and end of the pipe.

vii. **Jointing of the pipes:**
Pipe are laid properly by the helps of joints. It must be ensured to place valve at the proper place.

viii. **Testing of pipes:**
After laying and jointing, pipes are tested for non-leakage along the section of 500 m in length. One end of the pipe is closed and from another end, water is pumped with double of working pressure and kept for 24 hours. It is satisfactory if the leakage is not greater than the permissible limits.

Composting is the biological process in which dewatered sewage sludge is mixed with organic materials where microorganisms decompose the organic matter to produce a stable, nutrient-rich compost. Since large population of Nepal are engaged in agriculture, composting is highly suitable. This method of sludge treatment is environmentally friendly and appropriate for municipalities and small towns.
Strength:

i. Simple technology and easy to operate and maintain.
ii. Low construction and operating cost.
iii. Produces valuable compost that improves soil fertility.
iv. Suitable for decentralized treatment systems.

Weakness:
i. Requires larger land area.
ii. Composting takes longer period of time from several weeks to months.
iii. Odor and flies nuisance may occur if poorly managed.
iv. Requires proper aeration, moisture control and regular turning

Incineration is the process of controlled combustion of dewatered sludge at high temperatures at about 850-1000 degrees, converting into ashes, gases and heat. Incineration is suitable for large metropolitan wastewater treatment plants, hospital and industrial sludge containing hazardous contaminants. However, due to sophisticated air control systems and high energy consumption, it is generally less suitable for Nepal.
Strength:
i. Incineration method of sludge disposal reduces the sludge volume by 90-95%.
ii. Destroys pathogens completely.
iii. Eliminated offensive odors after combustion.
iv. Rapid treatment process.

Weakness:
i. Very high capital and operating costs.
ii. Required skilled operators and sophisticated equipment.
iii. May produce harmful air pollutants.
iv. Less suitable for small municipalities and rural areas.

Among the two methods, I as a sanitary engineer would recommend:
- Composting as the primary sludge treatment method for small and medium municipalities as it is economical, simple to operate and environmentally sustainable and produces valuable organic fertilizers for agriculture.
- Incineration for large urban wastewater treatment plants, hospital sludge or industrial wastes where complete destruction of pathogens and significant volume reduction takes place.

12. What is the significance of sustainable waste management systems in public health engineering? Explore the different technologies used for waste treatment, disposal and recycling and their effects on community health and the environment? [4+6=10]

Answer:
Sustainable waste management systems are the techniques of transforming water from an endpoint to recoverable resource by using recycling and reusing principles. These systems help to minimize environmental impact of the wastages produced, conserve natural resources and reduce the greenhouse gas emissions. In Nepal, different techniques are being considered for the sustainable waste management systems. One major example is Guheshwori Sewage Treatment Plant where the organic wastes from residential and commercial buildings are treated up to permissible degree before its final disposal into Bagmati river. Other methods such as greywater management, rainwater harvesting etc. are also commonly adopted in Nepal.

Significance of sustainable waste management systems in public health engineering:
i. Protection of public health:
- Prevents the spread of diseases caused by improper waste disposal.
- Reduces breeding of disease vectors such as flies, mosquitoes and rodents.

ii. Environmental preservation:

- Sustainable waste management systems help in preventing the contamination of wastages and cause pollution of soil, surface water or underground.
- It also helps to conserve ecosystems and biodiversity.

iii. Conservation of resources:

- The systems encourage recycling and reusing of materials such as paper, glass and metals.
- It helps to conserve natural resources and reduce the demand of new raw materials.

iv. Reduction of Greenhouse gases emissions:

- The techniques such as composting reduces the methane emission from the landfills.
- Recycling saves energy and well as helps to reduce carbon emissions and carbon footprints.

v. Economic benefits:

- Sustainable waste management systems also help in generating employment in waste collection, recycling and compost productions.
- Reduces landfill costs.
- Produces valuable products such as compost, recycled materials and bigas.

vi. Maintaining sanitary conditions:

- It also helps to maintain clean and sanitized surroundings.
- It reduces foul odors and visual pollution.

Technologies that are used for waste treatment, disposal and recycling:

i. Composting:

- Composting is the aerobic biological decomposition of biodegradable organic matter in which putrescible organic matters are decomposed into stable end products, and converted into compost.
- Compost has high soil conditioning value which is suitable for agriculture and horticulture.
- The method helps in reducing disease vectors, produce safer water disposal and improves agricultural productivity.

- There are three methods of composting:

  a. Composting by trenching:

  - Trench are excavated and refuse is laid in the layers of 15 cm in the trench.
  - Bacterial decomposition of organic matter take place which increases the temperature up to 75 degrees.
  - After 4 to 6 months, solid waste completely stabilizes which can be used as manure.

  b. Open windrow composting:

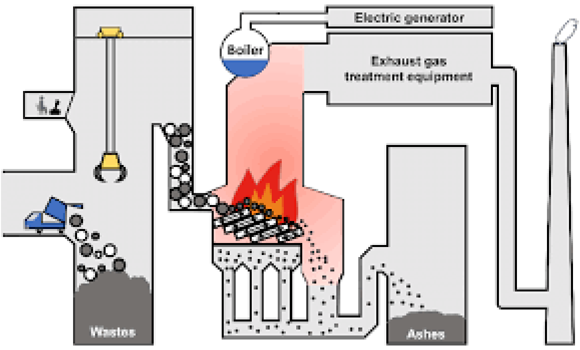
  - Refuse is laid in the form of piles or windrow where the bacterial decomposition of organic matter takes place.
  - The waste gets completely stabilized in 7 to 10 weeks.

  c. Mechanical composting:

- Piles of solid waste is turned by mechanical device.
- The stabilization of waste takes place in 1 to 2 weeks and can be used as compost.

ii. **Incineration:**

- It is one of the most hygienic and advanced method of solid waste management, suitable for combustible refuse.
- Refuse is burnt at high temperature **850–1200°C**, reducing waste volume by 80–90%.
- The method is suitable for medical wastes, hazardous waste and municipal wastes.
- The method helps in significant volume reduction, energy recovery and destroys pathogen.
- However, the capital cost is high and leads to air pollution if emission controls are inadequate.
- The method can be useful in energy generation but might cause air pollutions and carbon emissions affecting the respiratory health if not controlled properly.

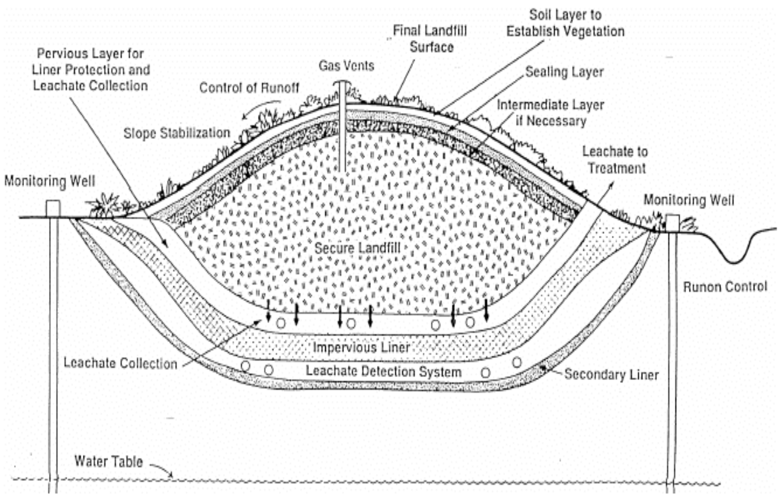


iii. **Vermicomposting:**

- It is the method of solid waste management where the organic waste is decomposed using earthworms.
- It produces high-quality compost, faster than conventional composting and is eco-friendly.
- However, the method is sensitive to temperature and moisture, suitable only for biodegradable waste.
- The method improves soil fertility and encourages organic farming.

iv. **Sanitary landfill:**

- It is an engineered facility for the safe disposal of solid waste with minimum environmental pollution and public health hazards.
- Sanitary landfill is characterized by the presence of a liner, leachate collection system that prevents groundwater contamination and a capping system that prevents air contamination.
- The method is safe for the disposal of residual waste and methane recovery is possible.
- However, long-term monitoring is required with the large land requirement.
- Poorly managed landfill sites may contaminate groundwater and create odor and pest problems.



v. **Recycling:**

- Materials are collected, sorted, processed and manufactured into new products.
- Recyclable materials include: paper, glass, plastics, aluminum, steel etc.
- It conserved natural resources, save energy and reduces landfill demand.
- However, recycling process requires source segregation and the contamination might lower the recycling efficiency.
- It helps in cleaner surroundings and reduced accumulation of wastes.

vi. **Pyrolysis:**

- Pyrolysis is the thermal decomposition of wastages that occurs in the absence of oxygen.
- It converts plastic waste into fuel and lowers emissions than open burning.
- However, the method is expensive and required advanced technology.
- The method reduces plastic pollution and recovers energy.